

WASP-52b

R-0533

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_240804 · created 2026-07-21T16:30:07+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460526.834 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.018
Transit depth (Rp/Rs)^2	2.73 +/- 0.61 [%]
Orbital Inclination (inc)	89.96 +/- 1.36
Airmass coefficient 1 (a1)	1.062 +/- 0.017
Airmass coefficient 2 (a2)	-0.048 +/- 0.02
Scatter in the residuals of the lightcurve fit is	1.62 %
Best Comparison Star	#1 - [282, 240]
Optimal Aperture	4.24
Optimal Annulus	17.48
Transit Duration (day)	0.0873 +/- 0.0021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.165 $\pm$ 0.018 vs 0.1646 (deviation 0.02 σ)
Duration fit vs published	0.0873 d vs 0.0758 d (deviation 4.43 σ)
Mid-time O-C	-8.0 min
Depth SNR	7.4
Residual scatter	1.62 %

Light curve

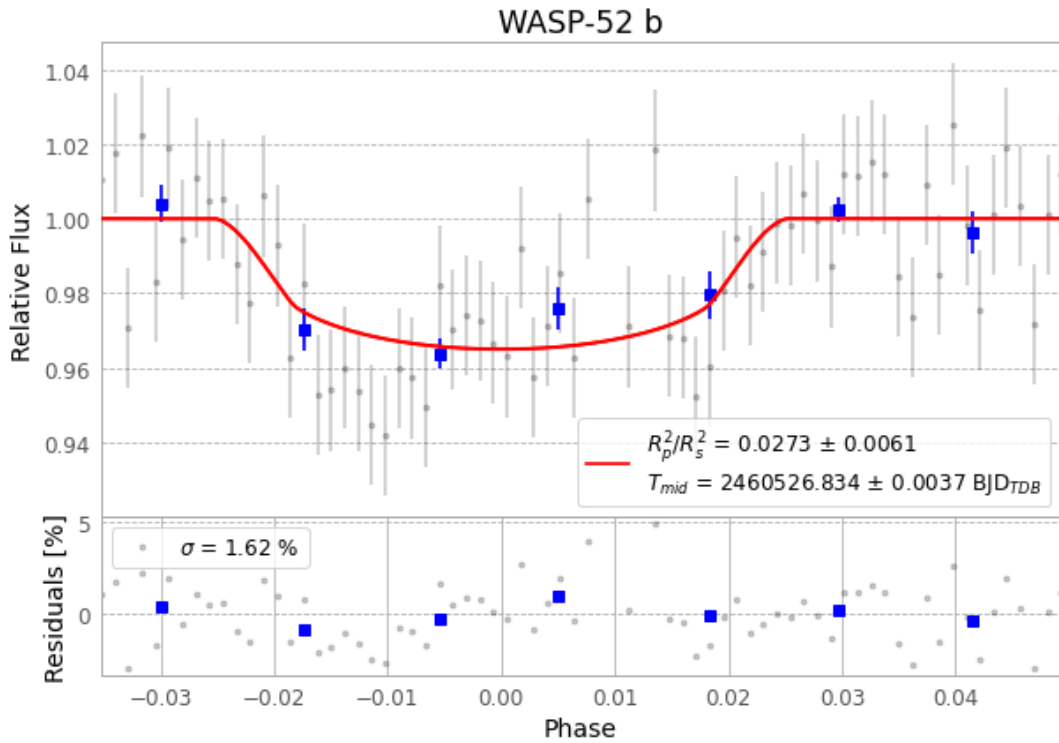


Figure 1 — FinalLightCurve_WASP-52 b_2024-08-03.png (SHA-256 1d88f5bf166d77b7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [307, 169], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f506c07cb36116b9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

77 FITS frames (51 MB), WASP-52240804060916.FITS → WASP-52240804095715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-240804.log (a9d1887c1281...), FinalLightCurve_WASP-52 b_2024-08-03.png (1d88f5bf166d...), FinalLightCurve_WASP-52 b_2024-08-03.csv (1bccbe50732d...)

Compute resources

Wall time 250.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-21 16:30Z.